Online Appendix to
Who Gains from Generative AI Model Releases?

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1 Sample Construction and Position Coding

The event universe combines public model chronologies with Artificial Analysis benchmark records. Same-day releases from the same model family are consolidated. Announcement dates are verified against vendor blogs, release notes, repositories, page metadata, and archived first captures. The pipeline retains 125 day-verified events and excludes one event whose identity is inconsistent across the two source systems. The return analysis uses 124 events. The annual counts are 3, 8, 45, 59, and 9 from 2022 through 2026.

The firm universe is the May 1, 2026 constituent list of the Nasdaq-100 Technology Sector Index. It contains 45 securities and 44 issuers. GOOG and GOOGL are retained as distinct traded securities and are combined at the issuer level in a robustness check. The balanced event-firm panel has 5,625 rows before the identity exclusion and regression-data requirements. The main position regression contains 5,441 observations, 124 events, and 45 securities.

Two coders independently evaluate 45 securities against 25 model developers. The resulting 1,125 firm–developer pairs contain eight binary position indicators. Table A1 reports agreement before adjudication. Overall agreement is 98.5% across 9,000 binary cells. Disagreements are resolved under the written codebook and retained in the decision files.

Table A1: Independent coding agreement

Position	Agreement (%)	Cohen's κ	Disagreements	Cells
Hardware supplier	100.0	1.000	0	1,125
Cloud platform	100.0	1.000	0	1,125
Downstream integrator	95.6	0.902	49	1,125
Downstream deployer	97.6	0.835	27	1,125
Downstream enabler	98.2	0.328	20	1,125
Competitor	96.7	0.823	37	1,125
Investor	100.0	1.000	0	1,125
Listed owner	100.0	1.000	0	1,125
All binary cells	98.5		133	9,000

Notes: Agreement is computed on the 45-security universe before adjudication. The enabler indicator is sparse, so its high raw agreement coexists with a lower prevalence-adjusted κ.

The capability sample contains 87 language-model events with comparable Artificial Analysis intelligence scores. The index has a standard deviation of 12.7 points. The remaining 37 events lack a comparable intelligence score. A separate media-generation flag identifies 41 events; four multi-component releases contain both a scored language model and a media model.

2 Position Results Across Benchmarks and Fixed Effects

Table A2 reports the two positive positions across the full set of return benchmarks. QQQ removes broad Nasdaq returns. SOXX absorbs the performance of the semiconductor sector. Event fixed effects compare firms inside the same release. Hardware and competitor coefficients remain close to their baseline magnitudes in each specification.

Table A2: Hardware and competitor coefficients across return models

Specification	Position	Coef. (pp)	Event CR2 p	Event $\times$ firm p
S&P 500 market model	Hardware	3.476***	0.0017	0.0145
	Competitor	3.385***	0.0003	0.0222
Fama–French three-factor	Hardware	3.428***	0.0021	0.0159
	Competitor	3.371***	0.0003	0.0368
QQQ market model	Hardware	3.459***	0.0016	0.0143
	Competitor	3.372***	0.0003	0.0253
SOXX market model	Hardware	2.898***	0.0027	0.0200
	Competitor	3.182***	0.0007	0.0282
S&P 500 with event fixed effects	Hardware	3.364***	0.0023	0.0282
	Competitor	3.259***	0.0004	0.0408

Notes: All regressions contain the eight position indicators and firm controls. The first four specifications include year fixed effects. The last includes event fixed effects. Stars use event-level CR2 p-values: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Dropping one of the 20 hardware securities at a time produces coefficients between 2.59 and 3.69 pp. All 20 estimates are positive and significant under event-level CR2 inference. Dropping NVIDIA yields 3.32 pp ($p = 0.0028$). Dropping Sandisk produces the smallest coefficient, 2.59 pp ($p = 0.0117$).

3 Pairwise Position Contrasts

Table A3 tests equality between the hardware coefficient and the other ecosystem positions. The hardware–competitor difference is close to zero. The hardware coefficient exceeds the average downstream and average nonhardware coefficients under both return benchmarks and both inference methods.

Table A3: Pairwise contrasts with the hardware coefficient

	Market model			Three-factor model		
Hardware minus	Difference	Event p	Two-way p	Difference	Event p	Two-way p
Cloud platform	2.426**	0.046	0.072	2.257*	0.063	0.103
Integrator	2.160**	0.025	0.051	2.243**	0.020	0.048
Deployer	2.214**	0.013	0.004	2.162**	0.014	0.007
Enabler	3.224***	0.004	0.014	3.561***	0.002	0.008
Competitor	0.090	0.917	0.924	0.057	0.946	0.957
Downstream average	2.533***	0.004	0.009	2.655***	0.002	0.007
Nonhardware average	2.023**	0.012	0.022	2.056***	0.010	0.019

Notes: Differences are percentage points. The downstream average combines cloud, integrator, deployer, and enabler coefficients. The nonhardware average combines the other seven position coefficients. Stars use event-level CR2 p-values: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Holm adjustment treats the eight position coefficients as one testing family. The adjusted p-values are 0.0119 for hardware and 0.0021 for competitor. The remaining six adjusted p-values exceed 0.41.

4 Time Interactions

The time specification includes each of the eight position indicators, all eight interactions with a 2025–2026 indicator, firm controls, and year fixed effects. The eight post-2025 interactions are jointly significant under every return-and-inference combination. Market-model p-values are 0.0048 with event CR2 and 0.0148 with event-and-firm clustering. Three-factor p-values are 0.0017 and 0.00002.

Table A4 reports the formal difference between the hardware and competitor changes. The estimate is approximately 5 pp under both return models. All confidence intervals exclude zero.

Table A4: Hardware change minus competitor change after 2025

Return model	Inference	Difference	SE	95% CI	p
Market model	Event CR2	4.863***	1.468	[1.954, 7.773]	0.0013
	Event × firm	4.863**	2.039	[0.755, 8.972]	0.0214
Three-factor model	Event CR2	5.032***	1.519	[2.022, 8.043]	0.0013
	Event × firm	5.032**	2.117	[0.766, 9.299]	0.0219

Notes: Difference and standard error are in percentage points. The contrast is $\Delta\beta_{\text{hardware}} - \Delta\beta_{\text{competitor}}$, where each Δ is the coefficient on the position's interaction with the 2025–2026 indicator. Stars use the p-value in each row: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

5 Capability Specifications

Table A5 reports the capability regressions. The pooled slope is negative under both return models. The market-model slope is more negative among closed-source events. The capability–hardware interaction is positive under both return models and both inference methods.

Table A5: Capability slopes and hardware interactions

Return model	Sample or term	Coef.	Event p	Two-way p	Observations	Events
Market model	Capability, all events	-0.0588^{*}	0.071	0.082	3,842	87
	Capability, closed source	-0.1305^{***}	0.005	0.017	2,207	50
	Capability, open weight	-0.0670	0.247	0.215	1,635	37
	Capability, unrelated firms	-0.0814	0.623	0.507	174	87
	Hardware at mean capability	1.8527^{**}	0.046	0.068	3,842	87
	Capability × hardware	0.1536^{**}	0.045	0.051	3,842	87
Three-factor model	Capability, all events	-0.0802^{**}	0.030	0.030	3,842	87
	Hardware at mean capability	1.9497^{**}	0.038	0.056	3,842	87
	Capability × hardware	0.1623^{**}	0.034	0.039	3,842	87

Notes: Coefficients are percentage points per intelligence-index point, except for the hardware indicator. Capability has a standard deviation of 12.7 points. Interaction regressions center capability before forming the product term. Stars use event-level CR2 p-values: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

6 Competitor Composition Tests

Competitor observations cover AAPL, ADBE, GOOG, GOOGL, META, and MSFT. Table A6 checks share classes, overlapping labels, and fixed effects. Combining GOOG and GOOGL into one issuer cluster leaves the two-way p-value at 0.022. Dropping either share class or the entire Alphabet issuer produces the same coefficient.

Table A6: Competitor composition and specification checks

Specification	Coef. (pp)	Event CR2 p	Two-way p
Baseline market model	3.385^{***}	0.0003	0.022
Cluster GOOG and GOOGL as one issuer	3.385^{***}	0.0003	0.022
Drop GOOG	3.396^{***}	0.0003	0.022
Drop GOOGL	3.396^{***}	0.0003	0.022
Drop Alphabet issuer	3.468^{***}	0.0003	0.023
Exclude competitor–cloud overlaps	3.478^{***}	0.0003	0.023
Exclude competitor–integrator overlaps	4.632^{***}	< 0.001	0.007
Exclude competitor–deployer overlaps	3.005^{***}	0.007	0.074
Event and firm fixed effects	1.320	0.414	0.032

Notes: The final specification absorbs time-invariant firm-level positions and has a rank-deficient design matrix for four coefficients. The competitor coefficient remains estimable. Stars use event-level CR2 p-values: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

In six leave-one-competitor-security regressions, coefficients range from 3.00 to 4.28 pp. All six are significant under event CR2; five are significant under two-way clustering. Dropping AAPL produces the largest two-way p-value, 0.074. Issuer-level deletion gives the same pattern.

Deleting one event creator at a time produces 25 additional regressions. Competitor coefficients range from 3.11 to 3.85 pp. Every estimate is significant under event CR2 and two-way clustering. The two-way p-values range from 0.011 to 0.034.

7 Dependence and Overlapping Events

Table A7 applies calendar-HAC adjustments to release-level coefficients. Each release receives equal weight. Hardware estimates retain their sign and magnitude as the bandwidth increases. Competitor estimates also remain positive, with wider confidence intervals at longer bandwidths.

Table A7: Event-level calendar-HAC inference

Position	Coef. (pp)	Lag 0	Lag 5	Lag 10	Lag 20	Lag 40
Hardware	3.283	0.002	0.025	0.044	0.071	0.102
Competitor	2.118	0.019	0.072	0.100	0.124	0.142

Notes: Entries after the coefficient are two-sided p-values. Bandwidths are trading days.

A maximal set of non-overlapping twenty-day windows contains 29 events. Across 250 random maximal sets, the median hardware coefficient is 4.22 pp and the median competitor coefficient is 3.78 pp. Every draw produces positive coefficients for both positions. With event-and-firm clustering, 49.6% of hardware draws and 64.0% of competitor draws have $p < 0.05$; the corresponding shares with $p < 0.10$ are 82.0% and 86.0%.

Calendar-time portfolios give each trading day one observation. The hardware post-minus-pre daily alpha is 7.50 basis points, with Newey–West p-values of 0.039, 0.039, 0.056, and 0.092 at lags 5, 10, 20, and 40. The competitor post-minus-pre alpha is 4.66 basis points, with p-values from 0.218 to 0.308. These portfolio results complement the firm-event regressions by changing both event and calendar-day weights.

8 Abnormal Trading Volume

Table A8 reports abnormal trading volume around model releases. The dependent variable is log volume relative to its $[-200, -11]$ estimation-window mean, scaled by the estimation-window standard deviation. Hardware and competitor announcement-volume coefficients are statistically insignificant in the NDXT45 sample.

Table A8: Abnormal volume before and after releases

	Pre-event $[-10, -2]$			Announcement $[0, +1]$		
Position	Coef.	Event p	Two-way p	Coef.	Event p	Two-way p
Hardware supplier	-0.077^{*}	0.051	0.353	-0.007	0.882	0.943
Cloud platform	0.074	0.148	0.299	0.095	0.161	0.400
Downstream integrator	-0.041	0.291	0.584	-0.052	0.340	0.605
Downstream deployer	-0.112^{***}	0.008	0.075	-0.170^{***}	0.004	0.054
Downstream enabler	-0.333^{***}	< 0.001	< 0.001	-0.457^{***}	< 0.001	< 0.001
Competitor	0.012	0.792	0.914	0.103	0.108	0.432
Investor	-0.097	0.232	0.436	-0.063	0.603	0.757
Listed owner	-0.058	0.646	0.690	0.563^{***}	0.007	0.032

Notes: Regressions include all position indicators, firm controls, and event fixed effects. Coefficients are in estimation-window standard deviations. Stars use event-level CR2 p-values: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

9 Relationship Codebook

The coding unit is a firm–model-developer pair. Event indicators take the union across developers represented in the event. Multiple indicators can equal one for the same pair.

Hardware supplier. The firm earns material revenue from physical inputs used in model training or inference, including accelerators, CPUs, memory, networking equipment, servers, storage, semiconductor design tools, and fabrication equipment.

Cloud platform. The firm operates hyperscale computing infrastructure that rents training or inference capacity and hosts third-party foundation models.

Downstream integrator. The firm embeds third-party foundation models into its own software product or platform. The model is an input into a product sold under the listed firm's brand.

Downstream deployer. The firm uses generative AI inside a broader operating business. The firm's primary product is outside the foundation-model and AI-infrastructure markets.

Downstream enabler. The firm advises, implements, integrates, or operates AI systems for enterprise clients as a service.

Competitor. The firm develops foundation models in the same modality as the releasing developer. Modality matching distinguishes language, image, video, audio, and multimodal competition.

Investor and listed owner. The investor indicator records a financial stake in the releasing entity. The owner indicator marks a listed parent or publisher responsible for the released model.

10 Reproducibility Map

The canonical event-firm panel is `Analysis/processed/event_firm_panel.csv`. Main position and capability estimates are generated by `Analysis/scripts/frl_two_way_cluster_and_equivalence.R`. Pairwise contrasts are generated by `Analysis/scripts/frl_position_contrasts.R`. Time interactions and competitor checks are generated by `Analysis/scripts/frl_time_and_competitor_robustness.R`. Non-overlapping samples and calendar-time portfolios are generated by `Analysis/scripts/frl_nonoverlap_calendar_portfolio.R`. The root entry point `run_reproduction.sh` rebuilds the source-derived data, the event-firm panel, every reported result, both coding validations, the figure, and both PDFs.